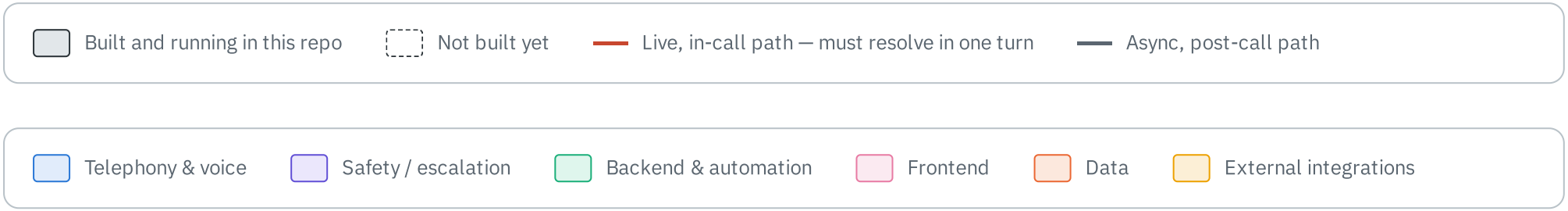
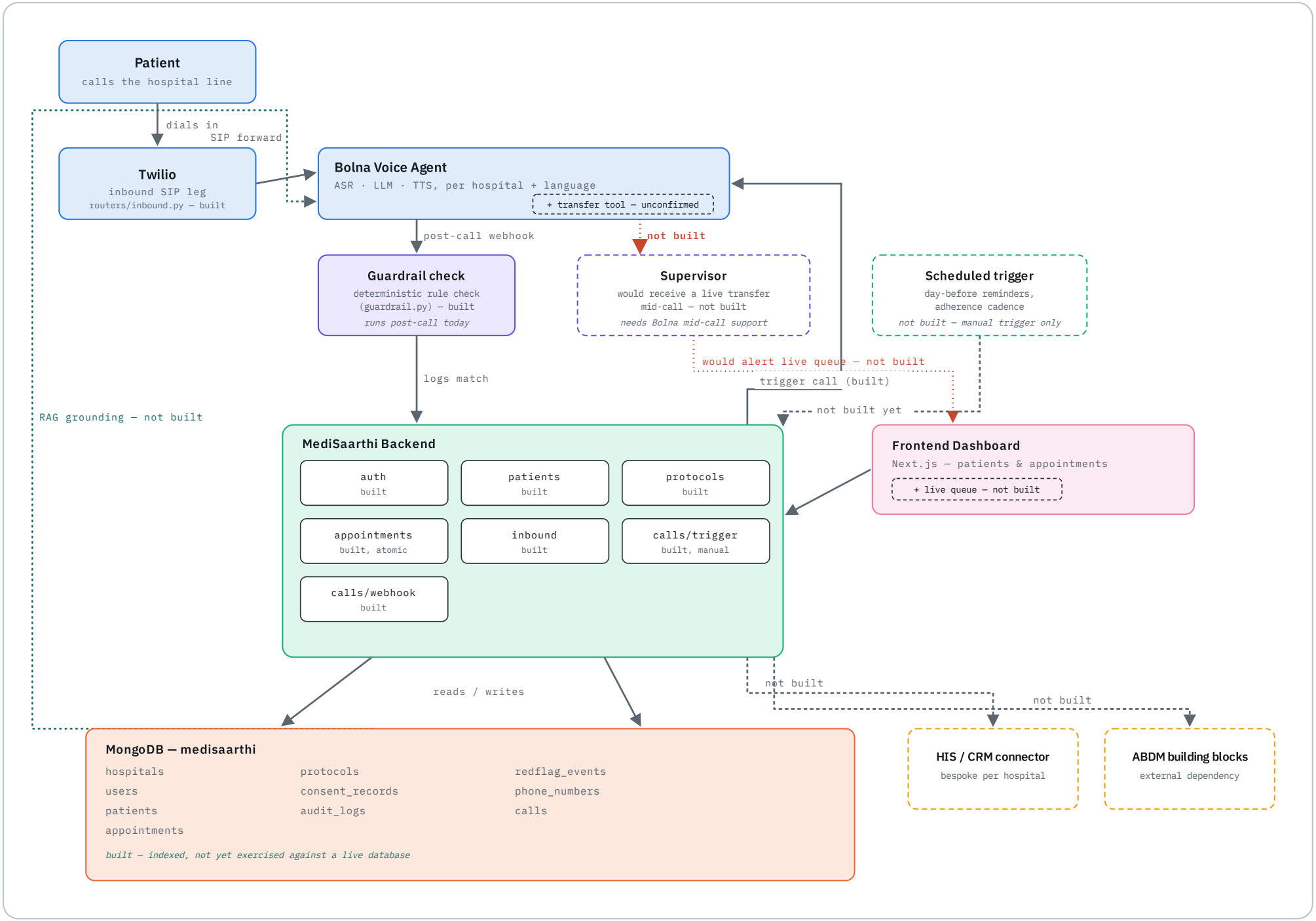
MEDISAARTHI · SYSTEM ARCHITECTURE

How a call becomes a care path

Every box below is either built and running in this repo today, or it isn't yet — the border style says which. The red path is the one that has to resolve inside a single conversation turn; everything else can afford to be a webhook. This is MediSaarthi's own independent backend — nothing here is imported from Talkly.




Patient
calls the hospital line
dials in
SIP forward
Twilio
inbound SIP leg
routers/inbound.py — built
Bolna Voice Agent
ASR · LLM · TTS, per hospital + language
+ transfer tool — unconfirmed
post-call webhook
not built
Guardrail check
deterministic rule check (guardrail.py) — built
runs post-call today
Supervisor
would receive a live transfer mid-call — not built
needs Bolna mid-call support
Scheduled trigger
day-before reminders, adherence cadence
not built — manual trigger only
would alert live queue — not built
trigger call (built)
not built yet
logs match
RAG grounding — not built
MediSaarthi Backend
auth
built
patients
built
protocols
built
appointments
built, atomic
inbound
built
calls/trigger
built, manual
calls/webhook
built
Frontend Dashboard
Next.js — patients & appointments
+ live queue — not built
reads / writes
not built
not built
MongoDB — medisaarthi
hospitals
users
patients
appointments
protocols
consent_records
audit_logs
redflag_events
phone_numbers
calls
built — indexed, not yet exercised against a live database
HIS / CRM connector
bespoke per hospital
ABDM building blocks
external dependency

Solid, filled boxes are built and running in this repo today. Dashed, outline boxes — and dashed red arrows — are not built yet. The live mid-call transfer is the one part of the system that has to work while the patient is still on the line, and it's the one piece not yet confirmed as buildable.

The live path still isn't built

The guardrail check (`guardrail.py`) is real and runs today — but only from the post-call webhook. The live, mid-call transfer depends on Bolna exposing a mid-call transfer/function-call tool, which hasn't been confirmed. That's still the first thing to verify.

Tenant isolation was rebuilt, not imported

MediSaarthi has its own `auth.py` with the same pattern Talkly uses — a hospital is a tenant, every query scoped by `hospital_id` — but it's fresh code in this repo, not a shared import. No dependency on Talkly at runtime.

RAG grounding isn't built yet

Protocols are stored (`protocols` collection, including each hospital's red-flag phrase list), but nothing retrieves them into a live Bolna call yet. Until that loop exists, "never diagnoses" is enforced only by what's in the prompt, not by what the agent can actually see.

`docs/architecture.html — built/not-built split verified against the MediSaarthi repo (../MediSaarthi/backend), not the pitch deck.`